The minors and cofactors of $a_{11}$, $a_{12}$, $a_{13}$ in a third order determinant $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ are as follows

(i) Minor of $a_{11}$ is $M_{11}$

$= \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22}a_{33} - a_{32}a_{23}$

Cofactor of $a_{11}$ is $A_{11}$

$= (-1)^{1+1}M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22}a_{33} - a_{32}a_{23}$

(ii) Minor of $a_{12}$ is $M_{12}$

$= \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = a_{21}a_{33} - a_{31}a_{23}$

Cofactor of $a_{12}$ is $A_{12}$

$= (-1)^{1+2}M_{12} = -\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$

$= -(a_{21}a_{33} - a_{31}a_{23})$

(iii) Minor of $a_{13}$ is $M_{13}$

$= \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} = a_{21}a_{32} - a_{31}a_{22}$

Cofactor of is $A_{13}$

$= (-1)^{1+3}M_{13} = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} = a_{21}a_{32} - a_{31}a_{22}$

## Example 1.2

Find the minor and cofactor of all the elements in the determinant $\begin{vmatrix} 1 & -2 \\ 4 & 3 \end{vmatrix}$

### *Solution*

Minor of 1 $= M_{11} = 3$

Minor of $-2$ $= M_{12} = 4$

Minor of 4 $= M_{21} = -2$

Minor of 3 $= M_{22} = 1$

Cofactor of 1 $= A_{11} = (-1)^{1+1}M_{11} = 3$

Cofactor of $-2$ $= A_{12} = (-1)^{1+2}M_{12} = -4$

Cofactor of 4 $= A_{21} = (-1)^{2+1}M_{21} = 2$

Cofactor of 3 $= A_{22} = (-1)^{2+2}M_{22} = 1$

## Example 1.3

Find the minor and cofactor of each element of the determinant. $\begin{vmatrix} 3 & 1 & 2 \\ 2 & 2 & 5 \\ 4 & 1 & 0 \end{vmatrix}$

### *Solution*

Minor of 3 is $M_{11} = \begin{vmatrix} 2 & 5 \\ 1 & 0 \end{vmatrix} = 0 - 5 = -5$

Minor of 1 is $M_{12} = \begin{vmatrix} 2 & 5 \\ 4 & 0 \end{vmatrix} = 0 - 20 = -20$

Minor of 2 is $M_{13} = \begin{vmatrix} 2 & 2 \\ 4 & 1 \end{vmatrix} = 2 - 8 = -6$

Minor of 2 is $M_{21} = \begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix} = 0 - 2 = -2$

Minor of 2 is $M_{22} = \begin{vmatrix} 3 & 2 \\ 4 & 0 \end{vmatrix} = 0 - 8 = -8$

Minor of 5 is $M_{23} = \begin{vmatrix} 3 & 1 \\ 4 & 1 \end{vmatrix} = 3 - 4 = -1$

Minor of 4 is $M_{31} = \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} = 5 - 4 = 1$

Minor of 1 is $M_{32} = \begin{vmatrix} 3 & 2 \\ 2 & 5 \end{vmatrix} = 15 - 4 = 11$

Minor of 0 is $M_{33} = \begin{vmatrix} 3 & 1 \\ 2 & 2 \end{vmatrix} = 6 - 2 = 4$

Cofactor of 3 is $A_{11}$

$= (-1)^{1+1}M_{11} = M_{11} = -5$

Cofactor of 1 is $A_{12}$

$= (-1)^{1+2}M_{12} = -M_{12} = 20$

Cofactor of 2 is $A_{13}$

$= (-1)^{1+3}M_{13} = M_{13} = -6$

Cofactor of 2 is $A_{21}$

$= (-1)^{2+1}M_{21} = -M_{21} = 2$

Cofactor of 2 is $A_{22}$

$= (-1)^{2+2}M_{22} = M_{22} = -8$

Cofactor of 5 is $A_{23}$

$= (-1)^{2+3}M_{23} = -M_{23} = 1$



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